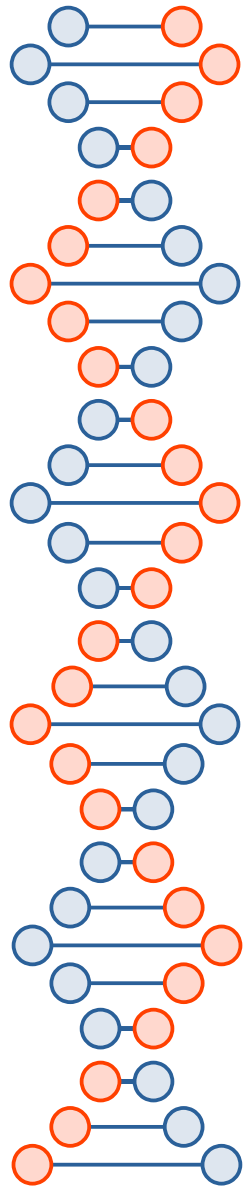


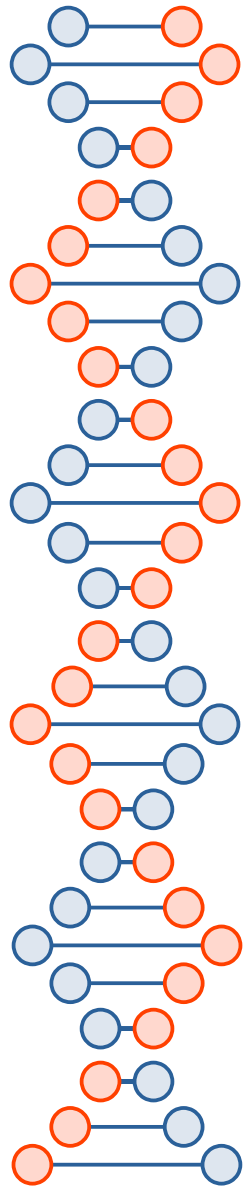
Myoglobin 1MBN Simulation

By – Krish Singh



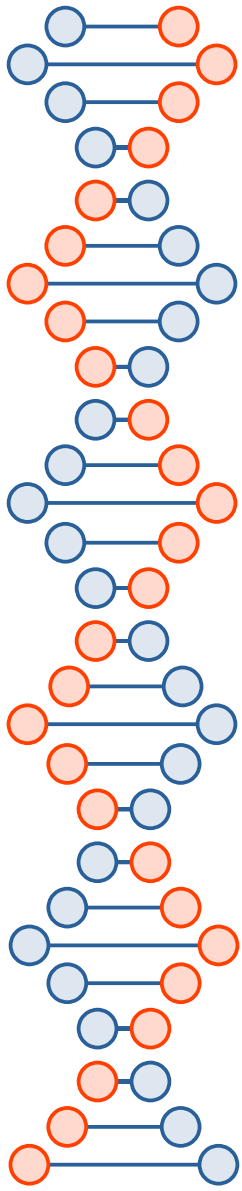
Objective

- To simulate protein in water
- To analyze structural stability

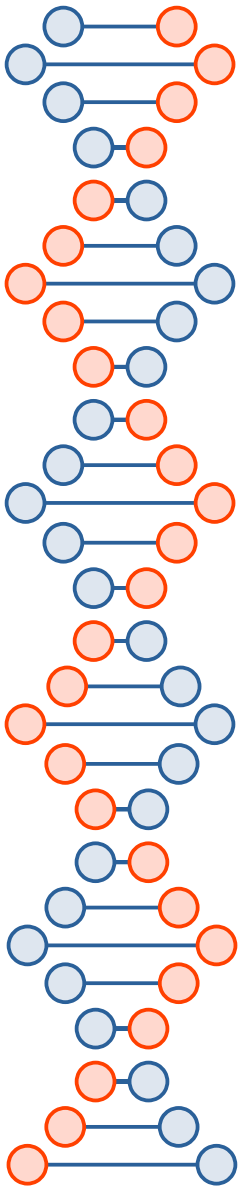


What was done ?

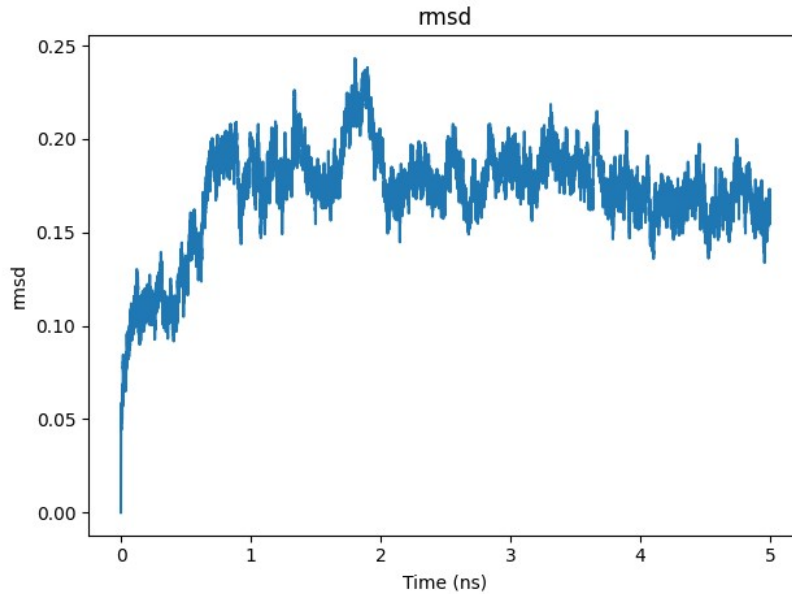
- Cleaned the structure (removed HETATM, water, etc.)
- Put it in a water box
- Added ions
- Minimized energy
- Ran equilibration
- Ran production MD
- Generated trajectory file (md.xtc)
- Analyzed that trajectory using GROMACS



OBSERVATIONS



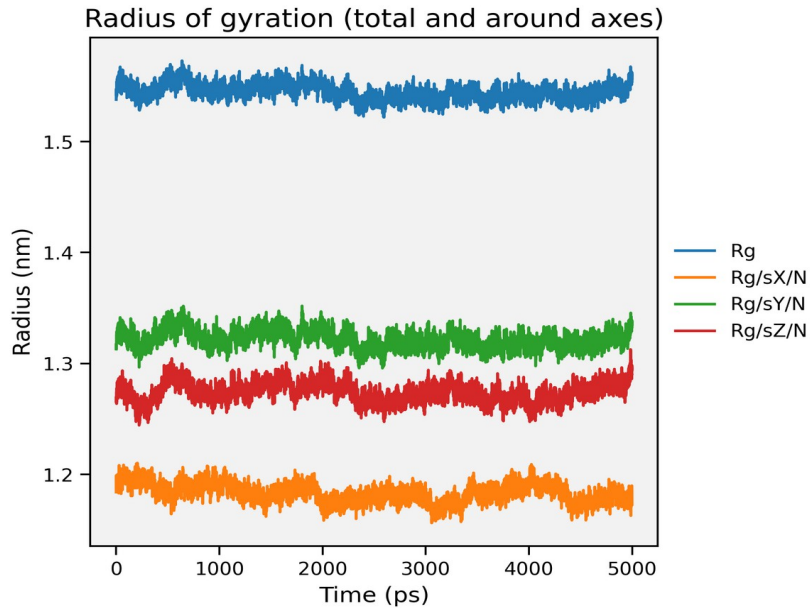
RMSD (Root Mean Square Deviation)



- The RMSD at frame 5000 was 0.1937267 nm, compared to 0.00 nm at frame 0.
- The RMSD increased initially due to structural relaxation and then stabilized, indicating that the system reached equilibrium and remained structurally stable.

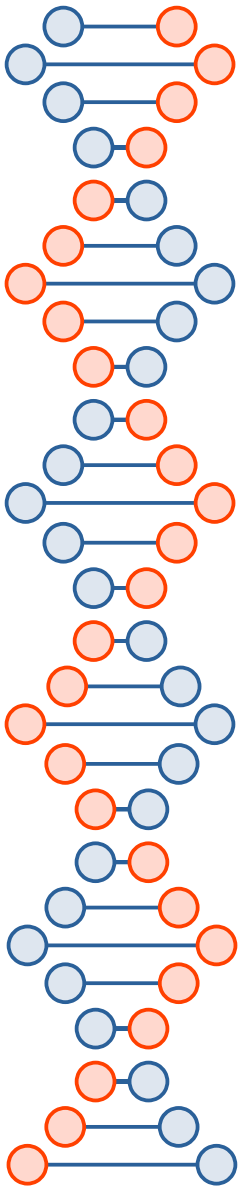
```
krish@krish-Predator-PHN16-71: ~/Desktop/Material-Informatics/MD Simula...  
(base) krish@krish-Predator-PHN16-71:~/Desktop/Material-Informatics/MD Simulation/  
Myoglobin 1MBN notebook$ grep -v '^[@#]' rmsd.xvg | sed -n '5000p'  
0.9998000    0.1937267  
(base) krish@krish-Predator-PHN16-71:~/Desktop/Material-Informatics/MD Simulation/  
Myoglobin 1MBN notebook$
```

Radius of Gyration (Rg)

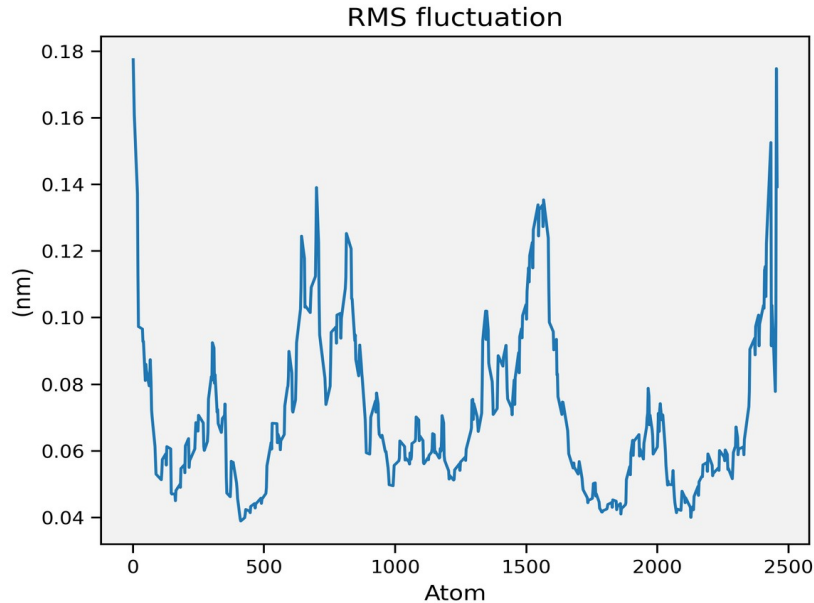


- The Radius of Gyration at frame 5000 was 1.543536 nm, compared to 1.537685 nm at frame 0.
- The Rg remained relatively constant, suggesting that the protein maintained its compact folded structure throughout the simulation.

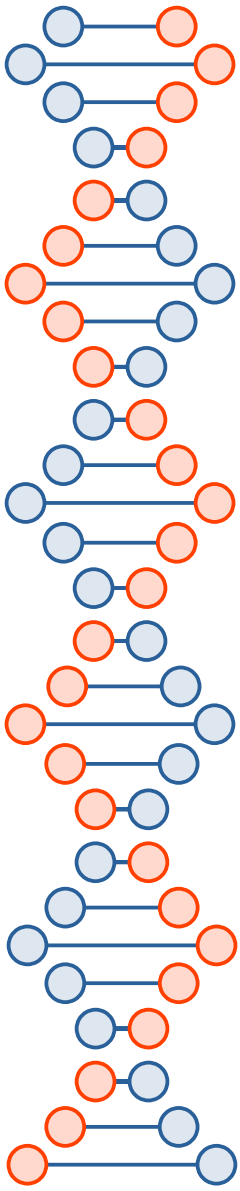
```
krish@krish-Predator-PHN16-71: ~/Desktop/Material-Informatics/MD Simul...  
(base) krish@krish-Predator-PHN16-71:~/Desktop/Material-Informatics/MD Simulation  
/Myoglobin 1MBN notebook$ grep -v '^[@#]' gyrate.xvg | sed -n '5000p'  
999.800 1.543536 1.181325 1.317567 1.278077  
(base) krish@krish-Predator-PHN16-71:~/Desktop/Material-Informatics/MD Simulation  
/Myoglobin 1MBN notebook$
```



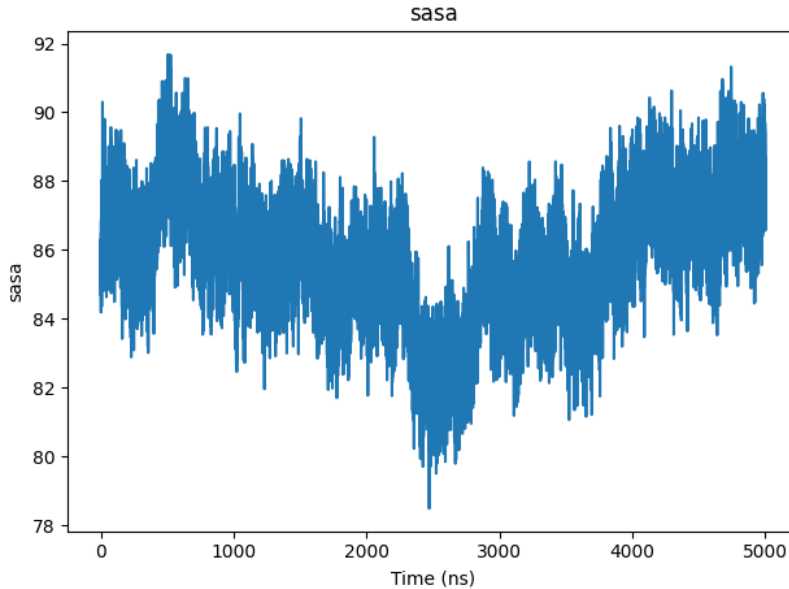
RMSF (Root Mean Square Fluctuations)



- Small values (~ 0.1 – 0.18 nm)
- Peaks at certain residues.
- Most atoms/residues are stable.
- Peaks = flexible loops or terminal regions.
- It shows Normal behaviour.
- High RMSF (>0.4 nm) is considered unstable.
- We don't have that.



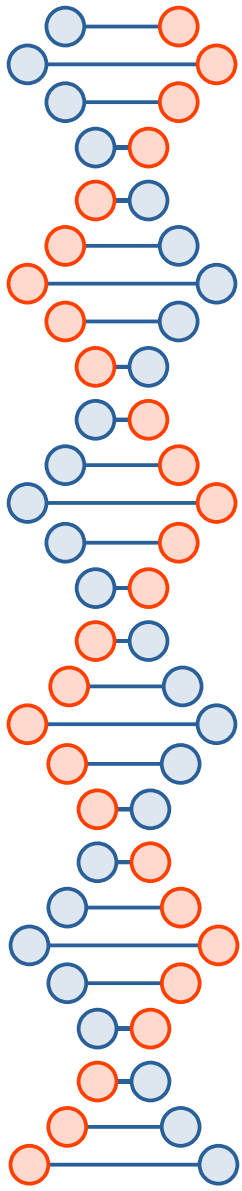
SASA (Solvent Accessible Surface Area)



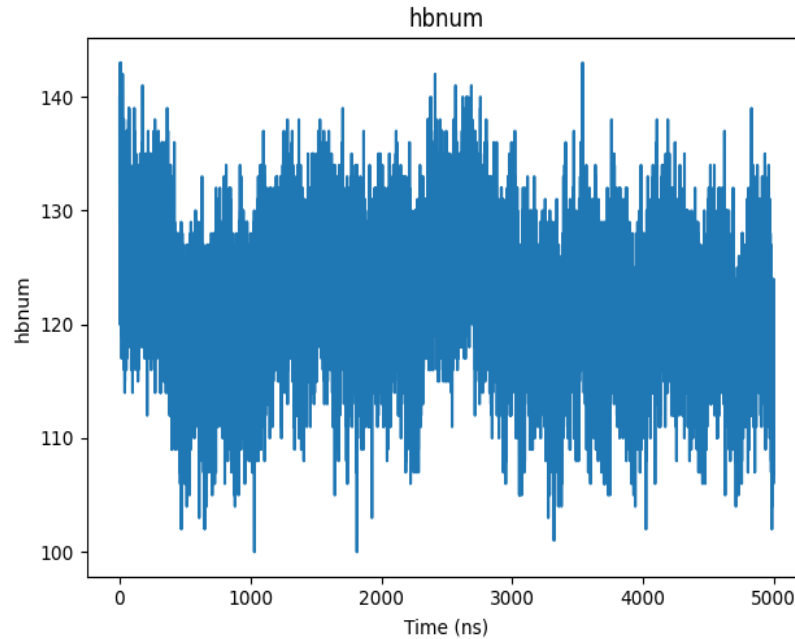
- Value around $\sim 85 \text{ nm}^2$ with small fluctuations.
- Structure not unfolding dramatically.
- No large exposure of hydrophobic core.
- System is structurally stable.

If protein unfolds SASA increases

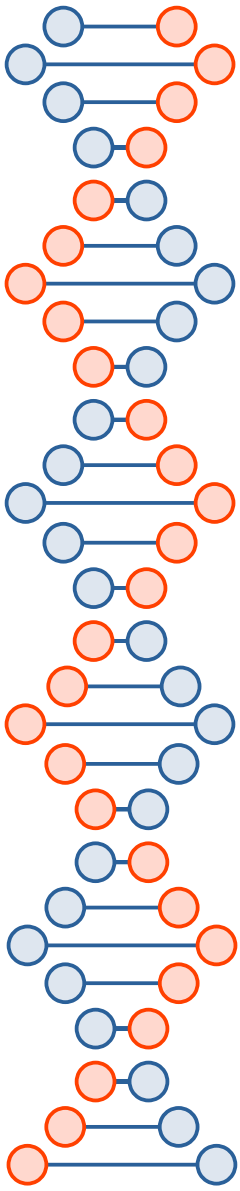
If protein stays folded SASA is stable



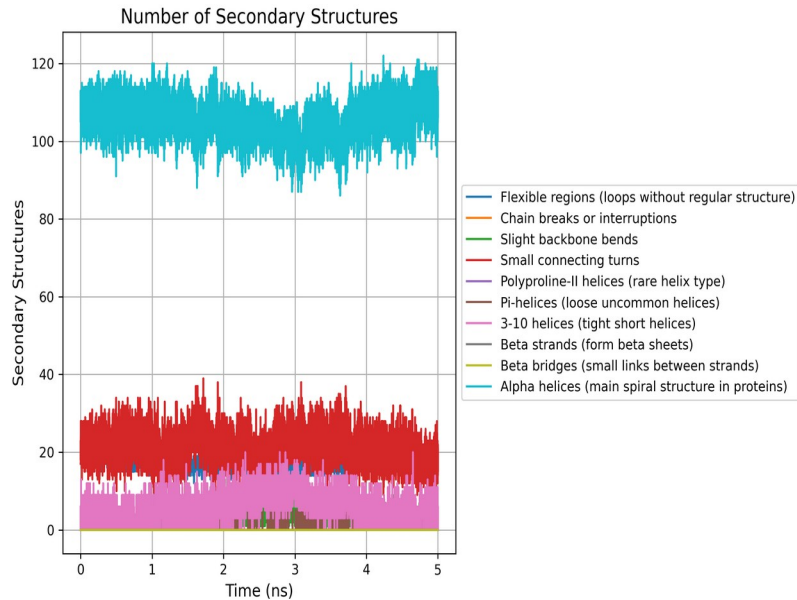
Hydrogen Bonds



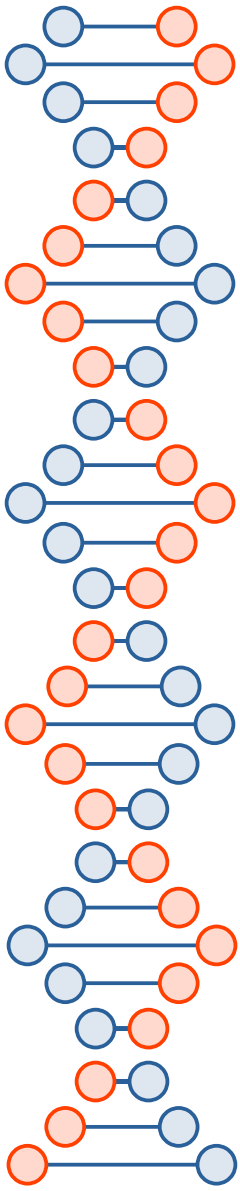
- Around 125–140 H-bonds fluctuating slightly.
- Hydrogen bond network is stable.
- No major structural collapse.
- Protein/polymer interactions are maintained.



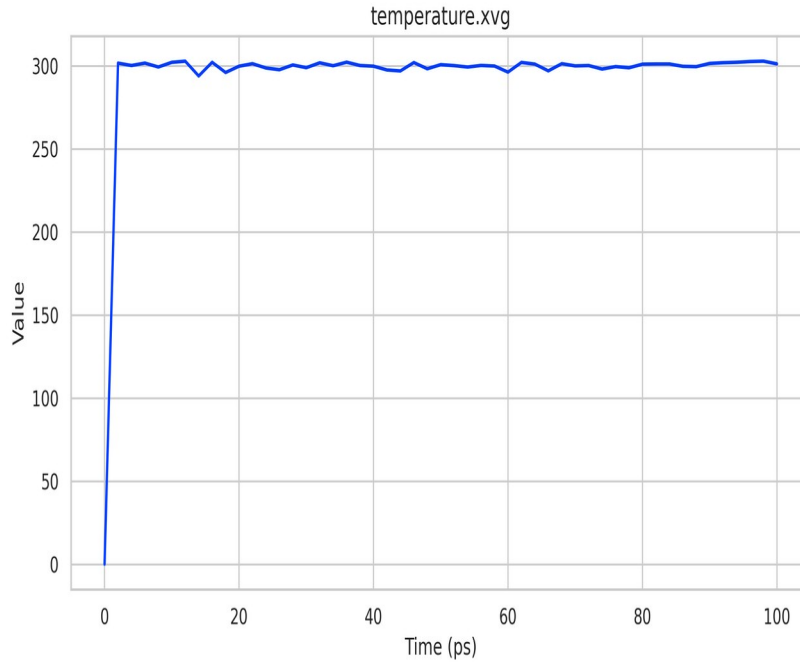
DSSP (Secondary Structure Evaluation)



- Alpha helices stay almost constant with only small changes.
- Turns and loops move a little, which is normal flexibility.
- Very few beta strands appear, so no major structure change happened.
- Overall secondary structure stayed stable during the whole simulation.

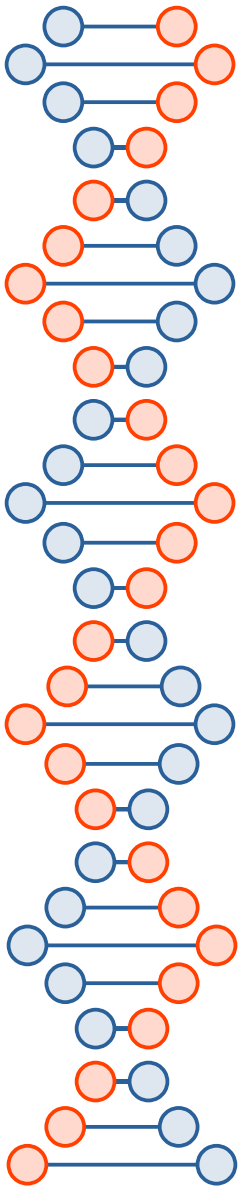


Temperature

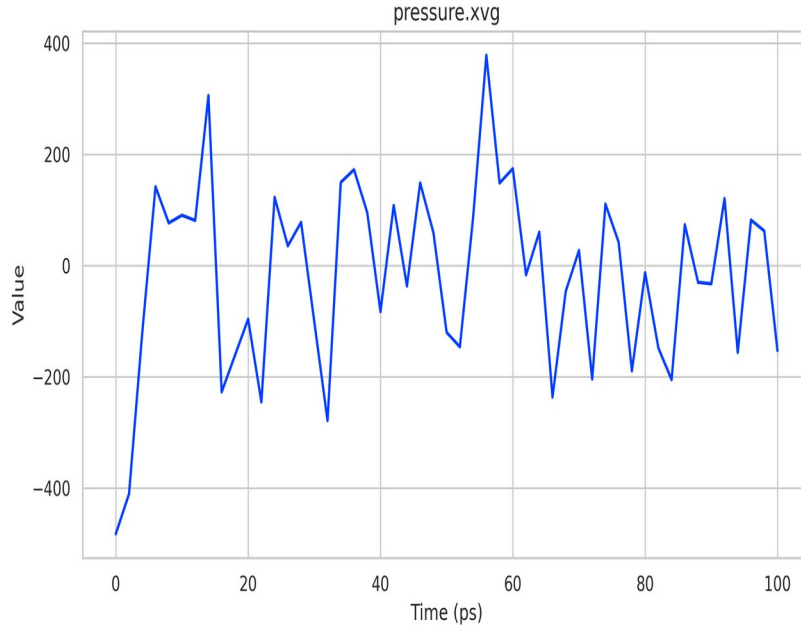


The temperature quickly reached the target value (~300 K) and remained stable with small fluctuations, indicating proper performance and thermal equilibration of the system.

- Temperature rapidly increases from ~0 K to ~300 K at the start.
- Stabilizes around the target temperature (~300 K).
- Minor fluctuations (≈ 296 – 303 K) are due to normal thermal motion.
- Confirms proper thermostat control and successful NVT equilibration.

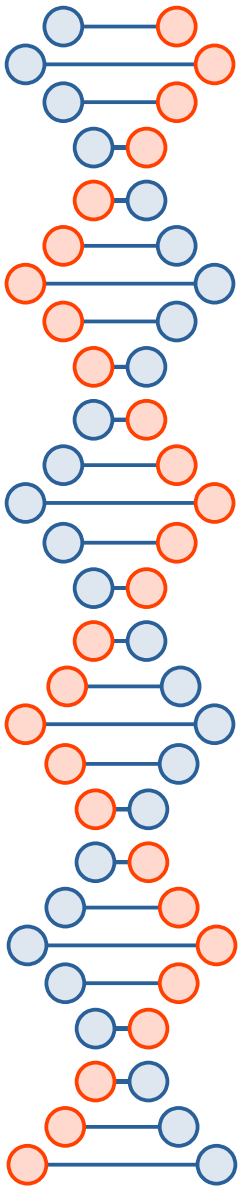


Pressure

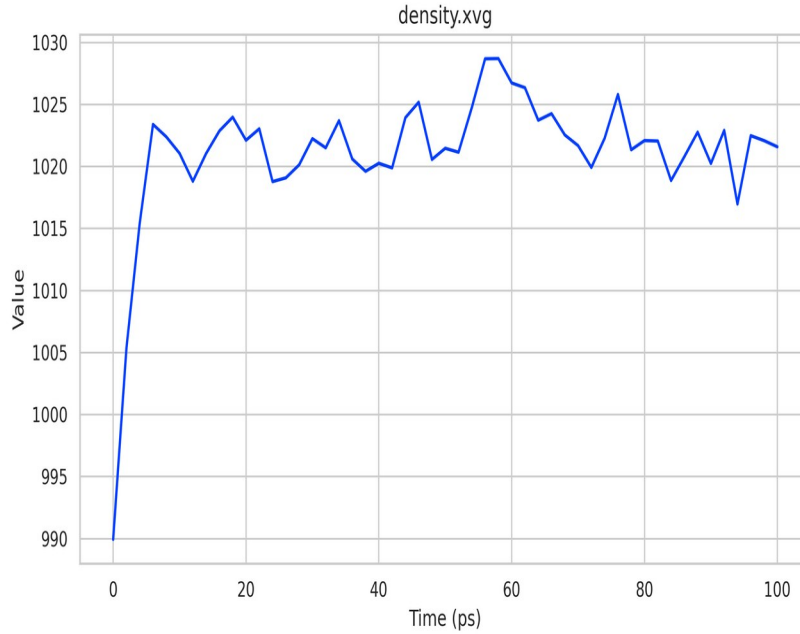


The pressure fluctuates widely between positive and negative values but oscillates around ~ 0 bar. This is normal during NPT equilibration and indicates the system is adjusting its volume to reach the target pressure.

- Pressure shows large fluctuations during equilibration phase.
- Values oscillate around the target pressure (~ 0 bar).
- Wide variations are normal for small MD systems.
- Confirms active pressure coupling and proper NPT equilibration.

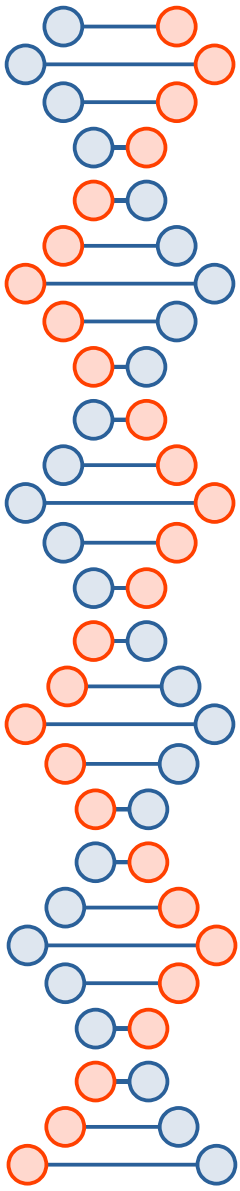


Density

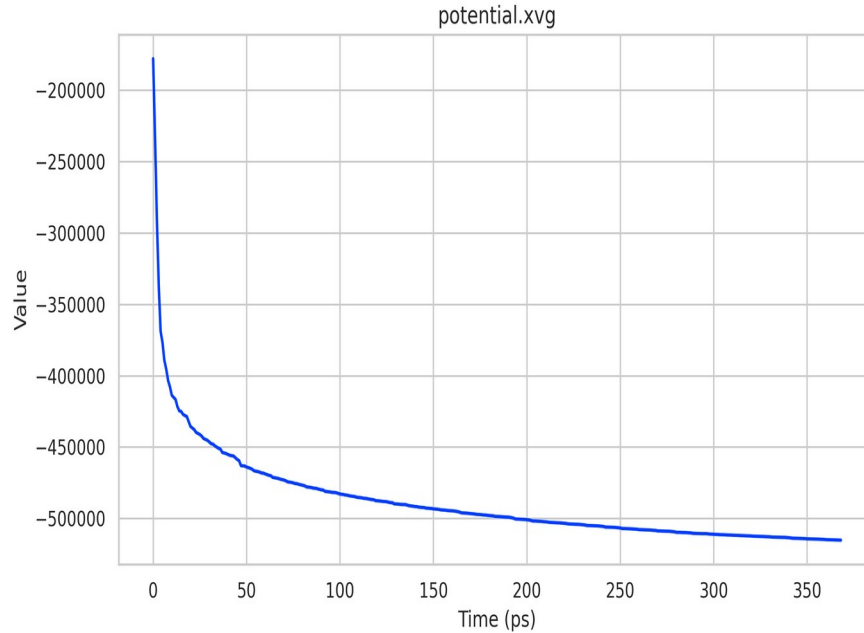


The density rises slightly at the start and then stays stable around $\sim 1020 \text{ kg/m}^3$. This shows the system reached equilibrium and the solvent box remained well-packed throughout the simulation.

- Initial rise indicates system equilibration during early simulation.
- Density stabilizes around expected water density ($\sim 1020 \text{ kg/m}^3$).
- Small fluctuations show normal thermal motion.
- Confirms proper and stable simulation.

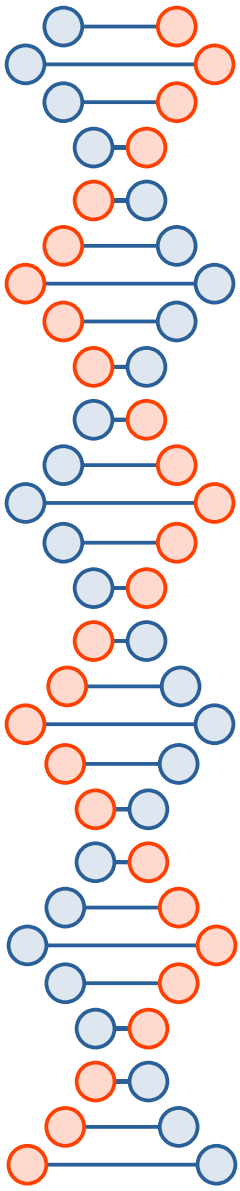


Potential Energy

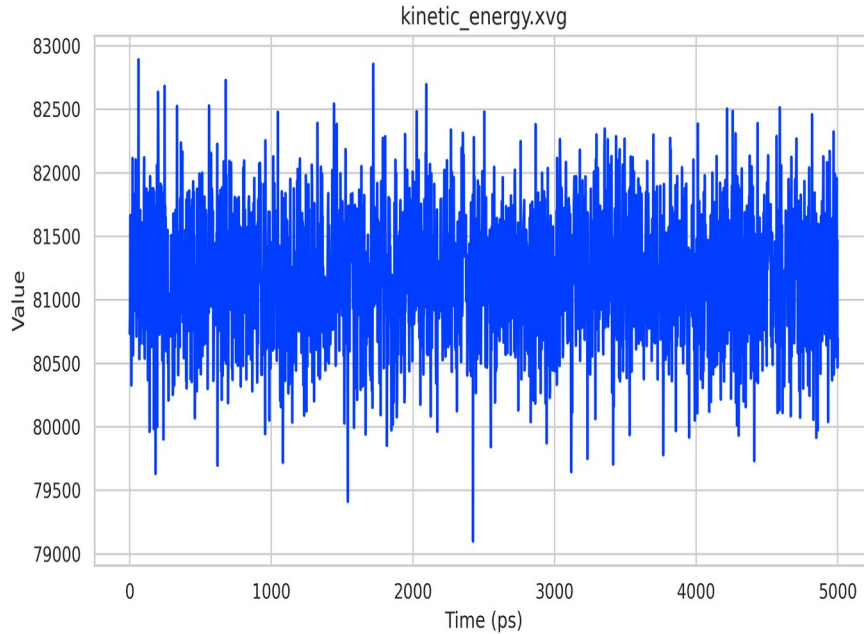


The potential energy steadily decreases from about -1.78×10^5 kJ/mol to around -5.15×10^5 kJ/mol, then begins to level off. This shows that the system is relaxing to a lower-energy, more stable configuration during energy minimization in GROMACS.

- Potential energy drops quickly at the beginning.
- Then it slowly starts to level off as the system settles.
- The more negative energy means bad contacts and clashes are being removed.
- When the curve becomes almost flat, it shows the structure has reached a stable state.

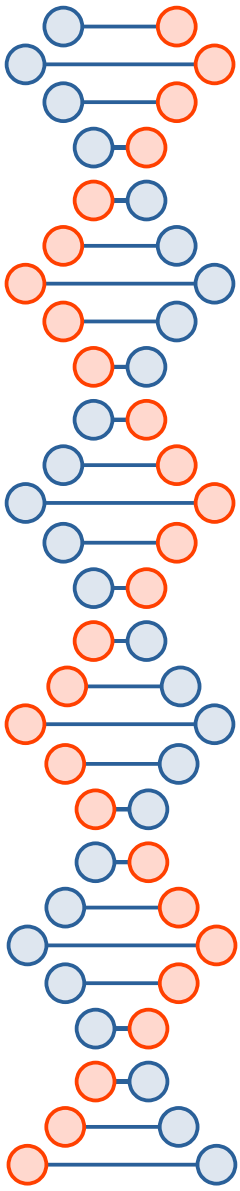


Kinetic Energy

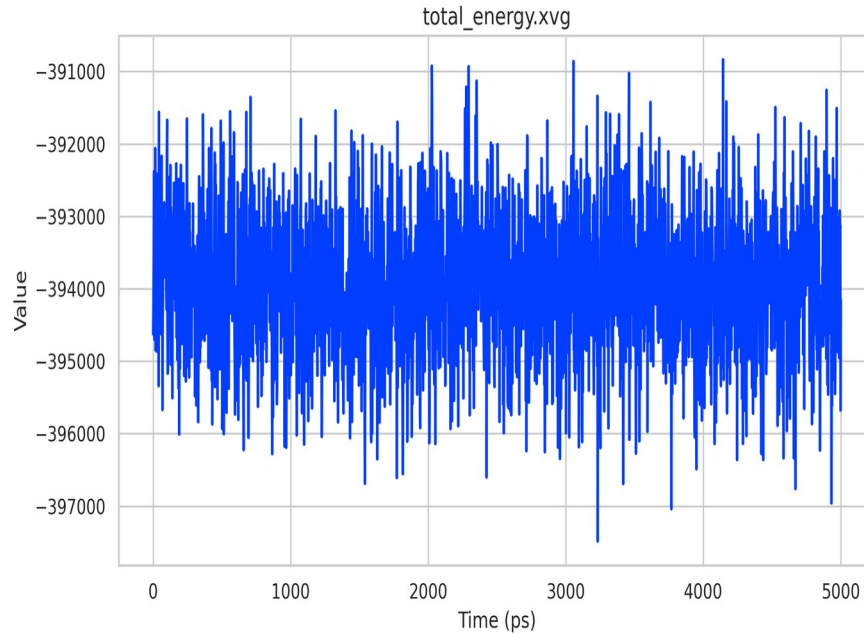


The kinetic energy fluctuates around $\sim 81,000$ kJ/mol throughout the simulation. The variations are small and random, indicating stable temperature and proper thermostat control.

- Kinetic energy fluctuates around $\sim 8.1 \times 10^4$ kJ/mol.
- No systematic increase or decrease over time is observed.
- Small fluctuations arise from normal thermal motion of atoms.
- Confirms stable temperature control and good equilibration of the system.

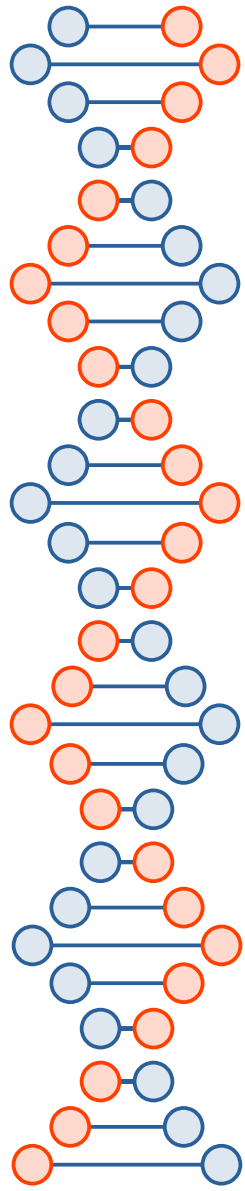


Total Energy



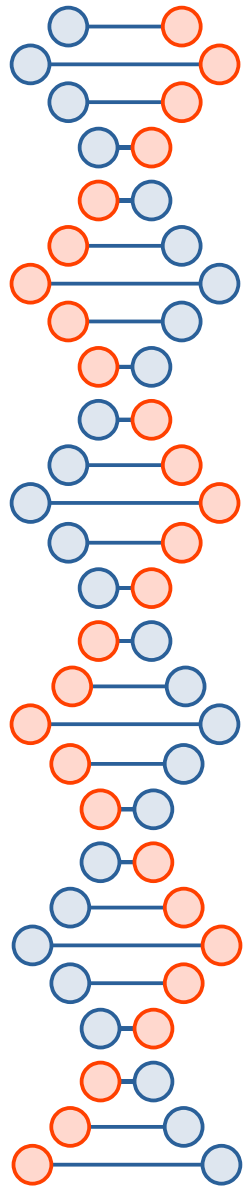
The total energy stays nearly constant around -3.94×10^5 kJ/mol with only small fluctuations. This shows the system remained stable and well equilibrated during the 5 ns run.

- Total energy remains stable around -3.94×10^5 kJ/mol.
- No noticeable drift indicates good energy conservation.
- Small fluctuations arise from normal thermal motion.
- Confirms the system is well equilibrated and simulation is stable.



Conclusion

All structural parameters remained stable.
The protein did not unfold during simulation. Therefore, the structure remained stable in the simulated aqueous environment.



References:

http://www.mdtutorials.com/gmx/lysozyme/01_pdb2gmx.html

<http://www.mdtutorials.com/gmx/lysozyme/index.html>